

by sharply reduced liquidity, and in some cases liquidity completely evaporated in some markets. Major illiquidity crises have occurred at least once every ten years, most in tandem with large downturns in asset markets.

2.3. SUMMARY

Illiquid asset classes as a whole are larger than the traditional liquid, public markets of stocks and bonds. Even normally liquid markets periodically become illiquid. Most investors' wealth is tied up in illiquid assets. Thus asset owners must consider illiquidity risk in the construction of their portfolios. Doing this requires estimating risk-return trade-offs of illiquid assets, but measuring illiquid asset returns is not straightforward.

3. Illiquid Asset Reported Returns Are Not Returns

As Faust and Forst note in their memo to Harvard's Council of Deans, the true illiquid asset losses were greater than the reported ones, which leads us to an important corollary. *Reported illiquid asset returns are not returns.* Three key biases cause people to overstate expected returns and understate the risk of illiquid assets:

1. Survivorship bias,
2. Infrequent sampling, and
3. Selection bias.

In illiquid asset markets, investors must be highly skeptical of reported returns.

3.1. SURVIVORSHIP BIAS

Survivorship bias results from the tendency of poorly performing funds to stop reporting. Many of these funds ultimately fail—but we only rarely count their failures. This makes true illiquid asset returns worse than the reported data.

Here's an analogy: Suppose we wanted to test the hypothesis that smoking is bad for you. We're going to run our tests only on a sample of smokers that have puffed cigarettes for at least twenty years and are in good health today. Lo and behold, we conclude that this *select* group of smokers has a slightly better mortality rate than the general population. Is this a valid conclusion? Of course not! We have taken a *biased sample* of smokers blessed with longevity who are, so far, invulnerable to the detrimental effects of tobacco. If you were to take up smoking today, what are the odds that you would end up in this lucky group twenty years later? Or would you die from emphysema (or heart disease, or lung cancer, etc.) before the experiment could be repeated in twenty years' time?